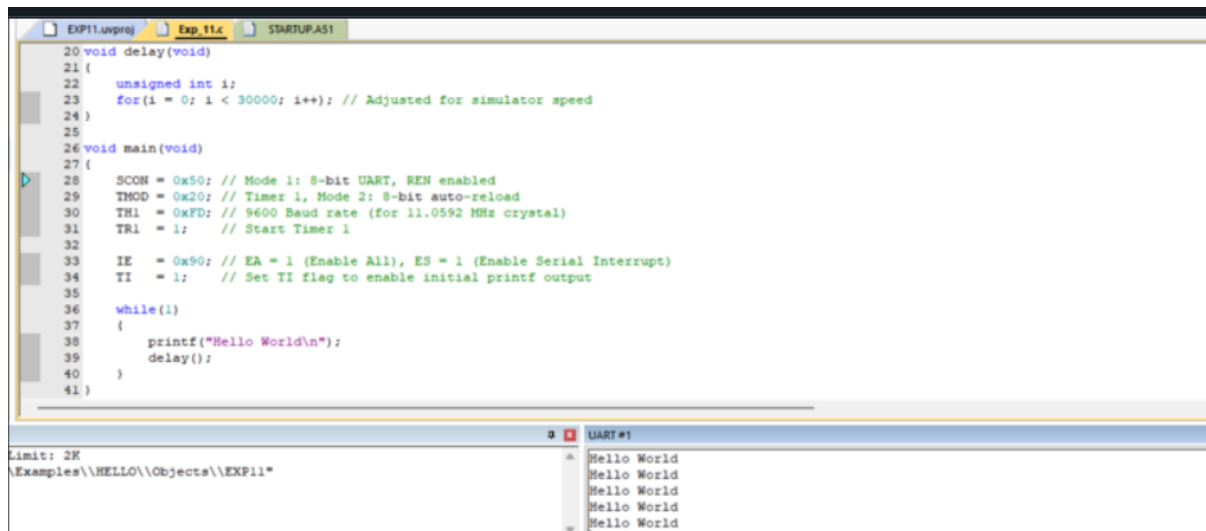


Exp No:16

SERIAL INTERRUPT USING UART (KEYBOARD INPUT) IN KEIL

AIM: To write and simulate an Embedded C program using serial interrupt in 8051 microcontroller such that while a message is continuously transmitted, any key pressed from the keyboard generates an interrupt and is displayed through UART



```
20 void delay(void)
21 {
22     unsigned int i;
23     for(i = 0; i < 30000; i++); // Adjusted for simulator speed
24 }
25
26 void main(void)
27 {
28     SCON = 0x50; // Mode 1: 8-bit UART, REN enabled
29     TMOD = 0x20; // Timer 1, Mode 2: 8-bit auto-reload
30     TH1 = 0x0FD; // 9600 Baud rate (for 11.0592 MHz crystal)
31     TR1 = 1; // Start Timer 1
32
33     IE = 0x90; // EA = 1 (Enable All), ES = 1 (Enable Serial Interrupts)
34     TI = 1; // Set TI flag to enable initial printf output
35
36     while(1)
37     {
38         printf("Hello World\n");
39         delay();
40     }
41 }
```

UART #1

Hello World
Hello World
Hello World
Hello World
Hello World

RESULT:

Thus, the serial interrupt using UART was successfully implemented and verified in Keil simulator. The interrupt was triggered upon keyboard input and serviced correctly using an Interrupt Service Routine.